ESP Registry

iir

Example of the component espressif/esp-dsp v1.3.0

To create a project from this example, run:

```
idf.py create-project-from-example "espressif/esp-dsp=1.3.0:iir"
```

or download archive (~3.40 KB)

IIR Filter Example

(See the README.md file in the upper level 'examples' directory for more information about examples.)

This example demonstrates how to use IIR filters functionality from esp-dsp library. Example does the following steps:

1. Initialize the library
2. Initialize input signal
3. Show LPF filter with Q factor 1
    - Calculate iir filter coefficients
    - Filter the input test signal (delta function)
    - Shows impulse response on the plot
    - Shows frequency response on the plot
    - Calculate execution performance
4. The same for LPF filter with Q factor 10

How to use example

Hardware required

This example does not require any special hardware, and can be run on any common development board.

Configure the project

Under Component Config ⟶ DSP Library ⟶ DSP Optimization, it's possible to choose either the optimized or ANSI implementation, to compare them.

Build and flash

Build the project and flash it to the board, then run monitor tool to view serial output (replace PORT with serial port name):

Plaintext

```
idf.py -p PORT flash monitor
```

(To exit the serial monitor, type `Ctrl-]`.)

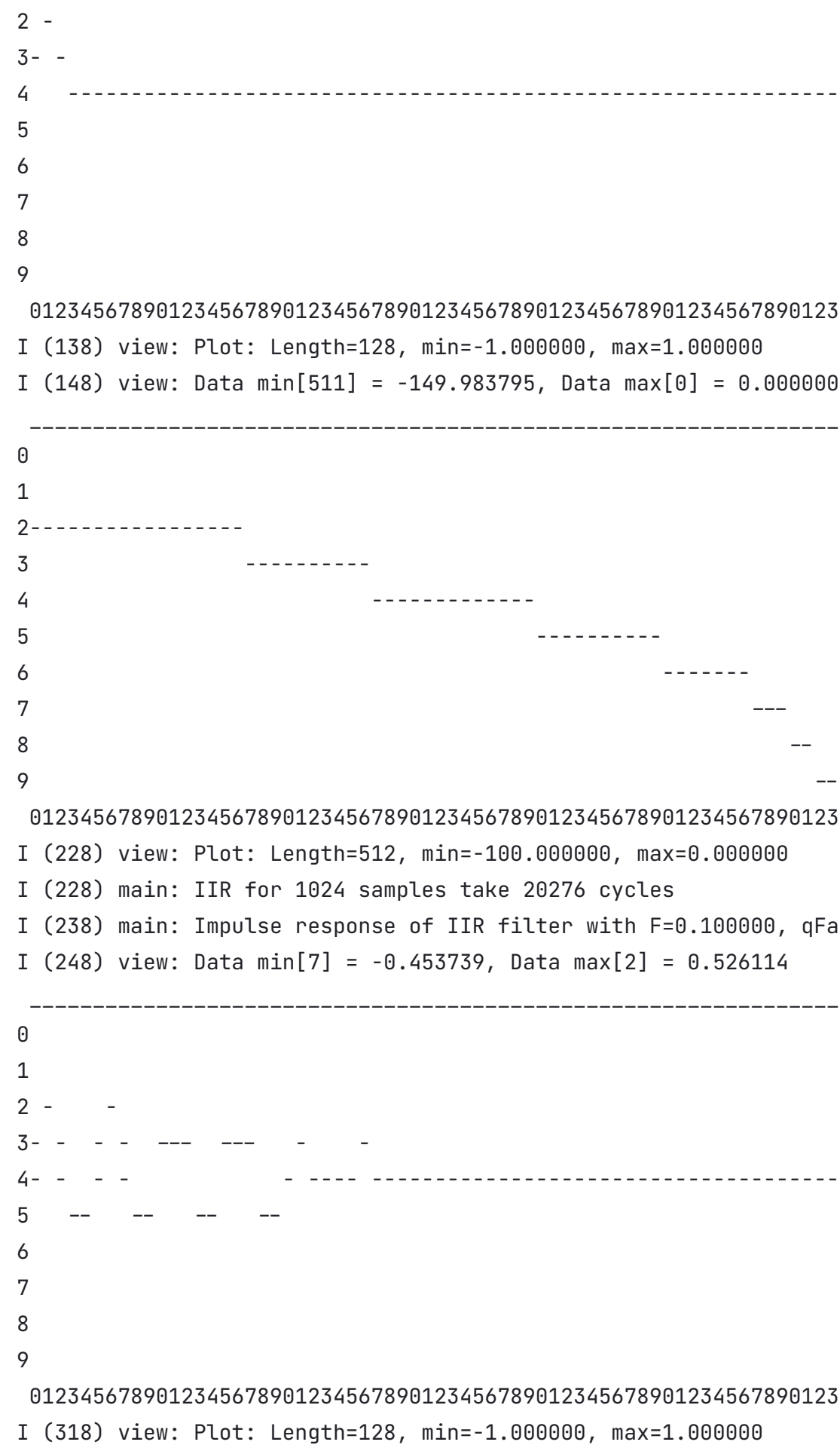
See the Getting Started Guide for full steps to configure and use ESP-IDF to build projects.

Example output

Here is an typical example console output.

Plaintext

```
I (58) main: Start Example.
I (58) main: Impulse response of IIR filter with F=0.100000, qFac
I (68) view: Data min[8] = -0.060052, Data max[2] = 0.333517
 ----------------------------------------------------------------
0
1
```


2 -
3- -
4 ----------
5
6
7
8
9
0123456789012345678901234567890123456789012345678901234567890123
I (138) view: Plot: Length=128, min=-1.000000, max=1.000000
I (148) view: Data min[511] = -149.983795, Data max[0] = 0.000000
0
1
2----------
3 ----------
4 ----------
5 ----------
6 -------
7 ---
8 --
9 --
0123456789012345678901234567890123456789012345678901234567890123
I (228) view: Plot: Length=512, min=-100.000000, max=0.000000
I (228) main: IIR for 1024 samples take 20276 cycles
I (238) main: Impulse response of IIR filter with F=0.100000, qFa
I (248) view: Data min[7] = -0.453739, Data max[2] = 0.526114
0
1
2 - -
3- - - - --- --- - -
4- - - - - ---- ----------
5 -- -- -- --
6
7
8
9
0123456789012345678901234567890123456789012345678901234567890123
I (318) view: Plot: Length=128, min=-1.000000, max=1.000000

```
I (328) view: Data min[511] = -149.480377, Data max[0] = 0.000000
 ------------------------------------------------------------
0            --
1          --  -
2----------     -----
3                    --------
4                            ------------
5                                        ----------
6                                                  -------
7                                                         ---
8                                                            --
9                                                              --
 01234567890123456789012345678901234567890123456789012345678901234567890123
I (408) view: Plot: Length=512, min=-100.000000, max=0.000000
I (408) main: IIR for 1024 samples take 17456 cycles
I (418) main: End Example.
```

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